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Database Normalization & Debugging Report

1. What was wrong

Normalization Issues: In the original database, multi-valued fields like tags and social media links (Instagram, Facebook) were stored as single strings in one column, violating 3rd Normal Form (3NF).

Syntax Errors: The initial code used SQL Server specific syntax (such as NVARCHAR, IDENTITY, and GETDATE()), which is incompatible with the MariaDB/MySQL engine used in XAMPP.

Lack of Relationships: There were no defined Foreign Key constraints, meaning the tables were isolated and data integrity could not be guaranteed.

Incorrect Data Types: Large text fields like blog content were improperly set as longblob instead of longtext, making data retrieval and display inefficient.

2. What I fixed

Syntax Standardization: Converted the entire script to standard MySQL/MariaDB syntax to ensure 100% compatibility with phpMyAdmin.

3NF Normalization: Restructured the database into 3rd Normal Form by creating dedicated tables for roles, tags, and user_socials to eliminate data redundancy.

Relationship Mapping: Established PRIMARY KEY and FOREIGN KEY constraints across all tables to link data logically.

Automation: Configured AUTO_INCREMENT for all IDs and implemented automatic timestamp fields for tracking entry and update times.

3. What improvements I added

Role-Based Access Control: Added a roles table to manage permissions for different user levels (Admin, Editor, User).

Many-to-Many Tagging System: Implemented a pivot table (blog_tags) allowing a single blog post to have multiple tags, which is essential for professional blogging platforms.

Referential Integrity: Added ON DELETE CASCADE rules. Now, if a user is deleted, their associated blogs and comments are automatically removed, preventing "orphan" data.

Feature Enhancements: Added a status column (Draft/Published) and a view counter to the blog table to support real-world content management workflows.



